

2025 Cloud Web Services: Course Introduction

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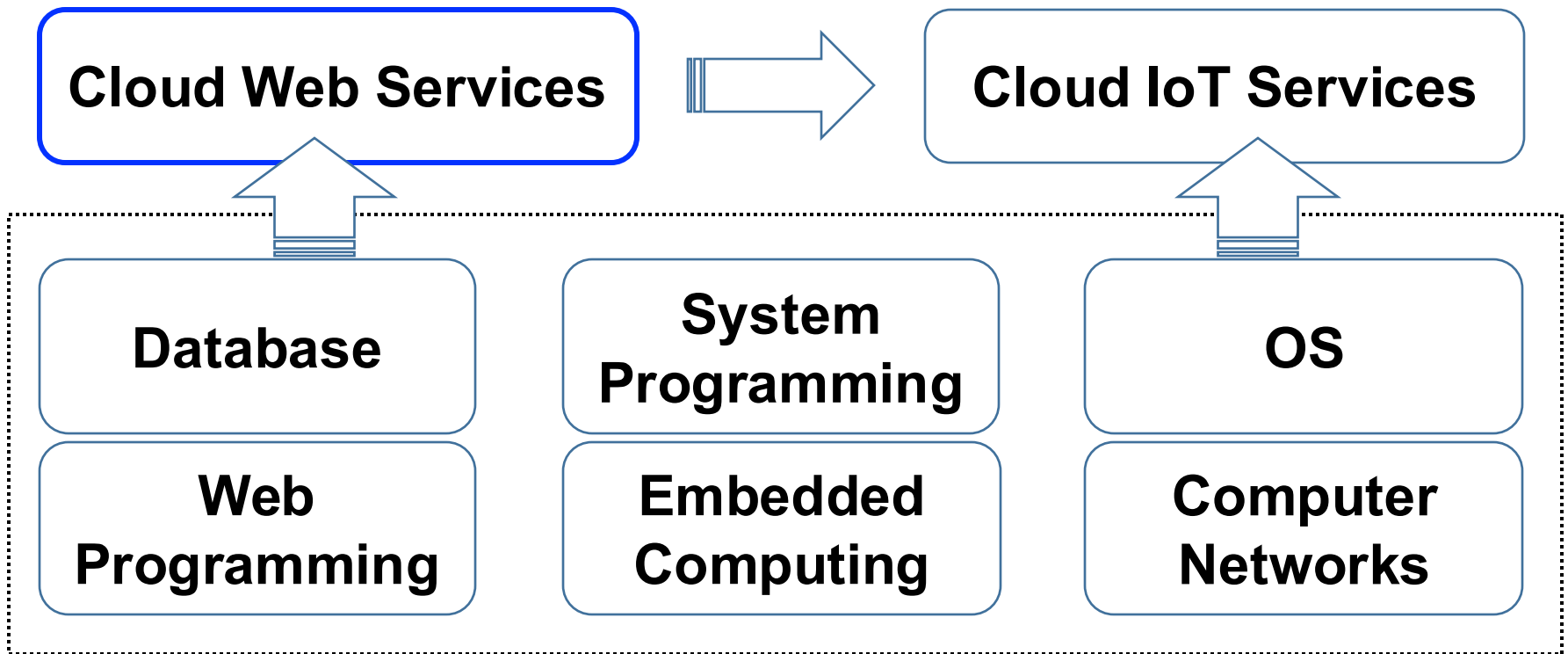
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Road Map

Architecture: ROA, SOA, Microservices
AWS: EC2, S3, Lambda, API Gateway, IAM
DB: NoSQL MongoDB

Architecture: EDA
AWS: IoT Core, SNS, EventBridge, IAM
Device: Raspberry Pi, Sensors



Cloud Web Services: Course Objectives

- First, we study the **concepts** of **Cloud Computing**, **RESTful Web Services**, **Serverless Computing**, and **Microservices**
- Second, we learn a **server development framework**: **Node.JS** and **Express.JS**
- Finally, we learn **how to use Amazon Cloud Services (AWS)** to build applications
 - Open Data Portal (<http://www.data.go.kr>)



Course Schedule (1/2)

Week	Topic
01	Introduction: Future IT Systems & Cloud Computing
02	Programming: NodeJS
03	Programming: NodeJS
04	Virtual Server: EC2
05	Virtual File: S3
06	Protocol: HyperText & HTTP
07	Architecture: REST Model, REST API, REST Framework (Express, Axios)
08	Q&A

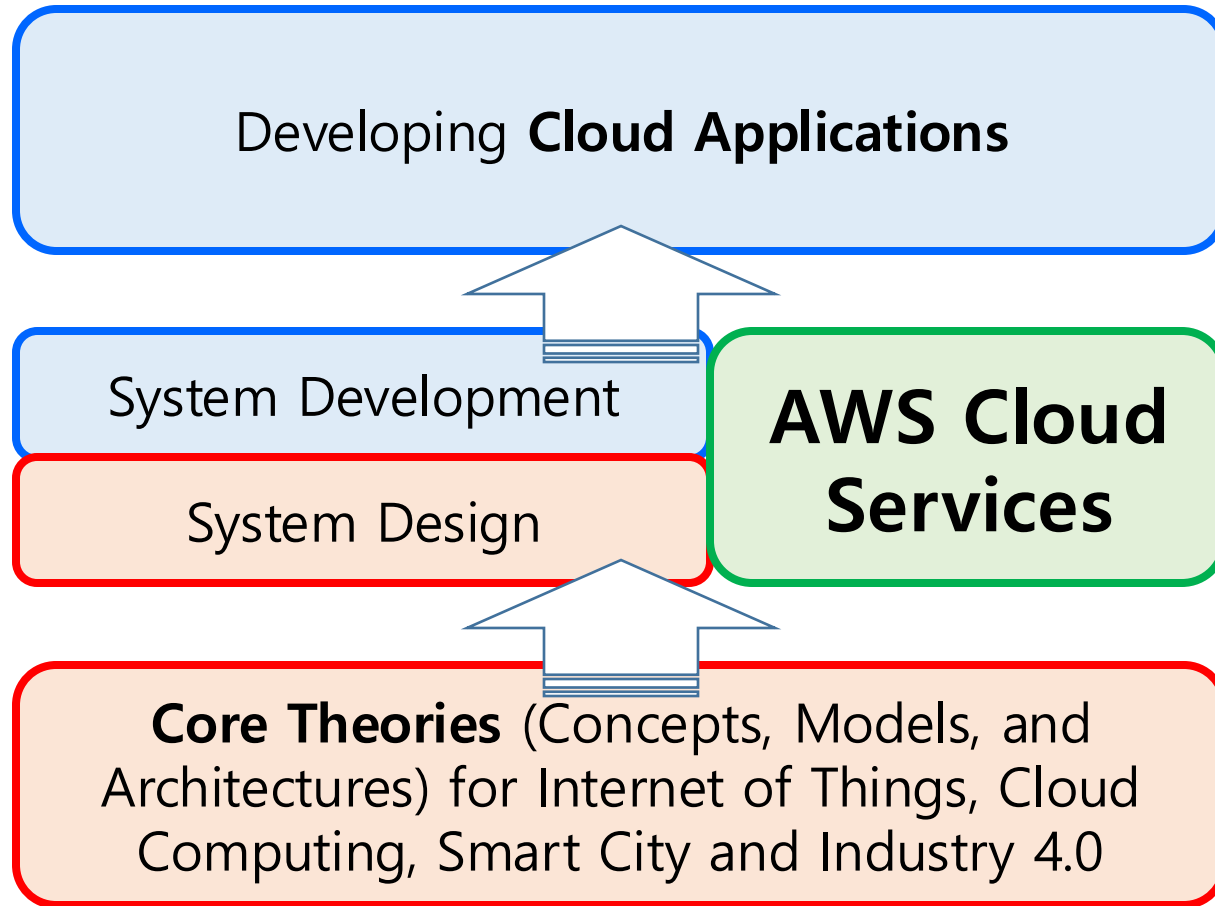


Course Schedule (2/2)

Week	Topic
09	Security: IAM #1
10	Security: IAM #2
11	Amazon Lambda & Serverless Computing
12	Amazon API Gateway
13	DBMS: MongoDB
14	DBMS: MongoDB
15	Project 결과 발표 및 평가
16	Q&A

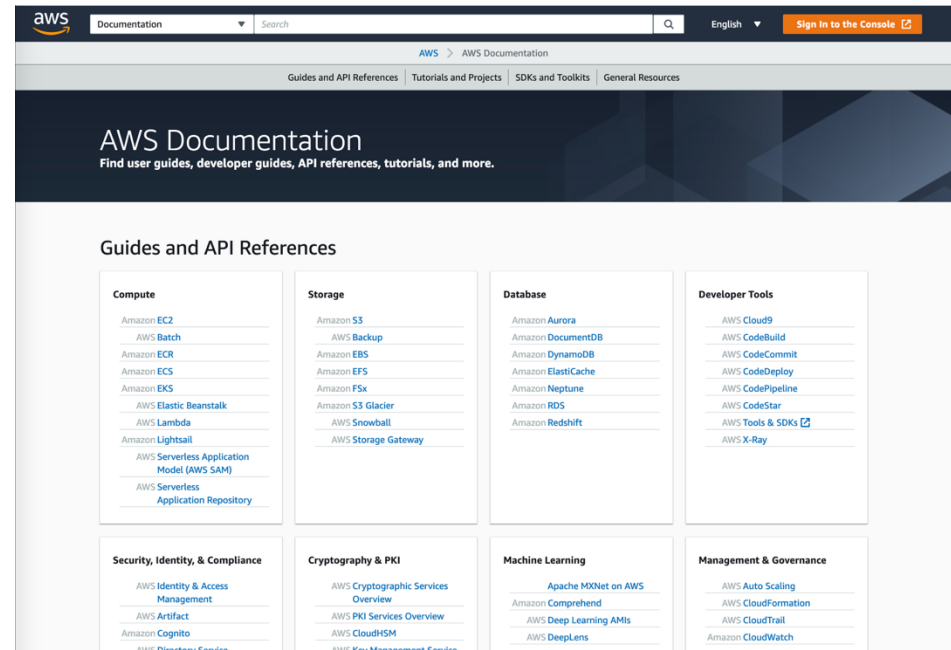
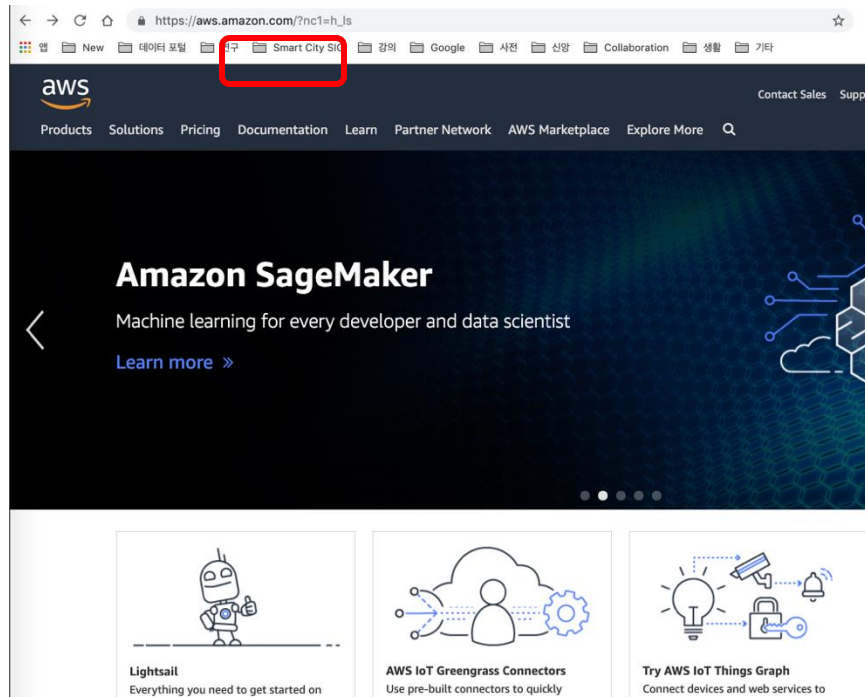


Course Objectives

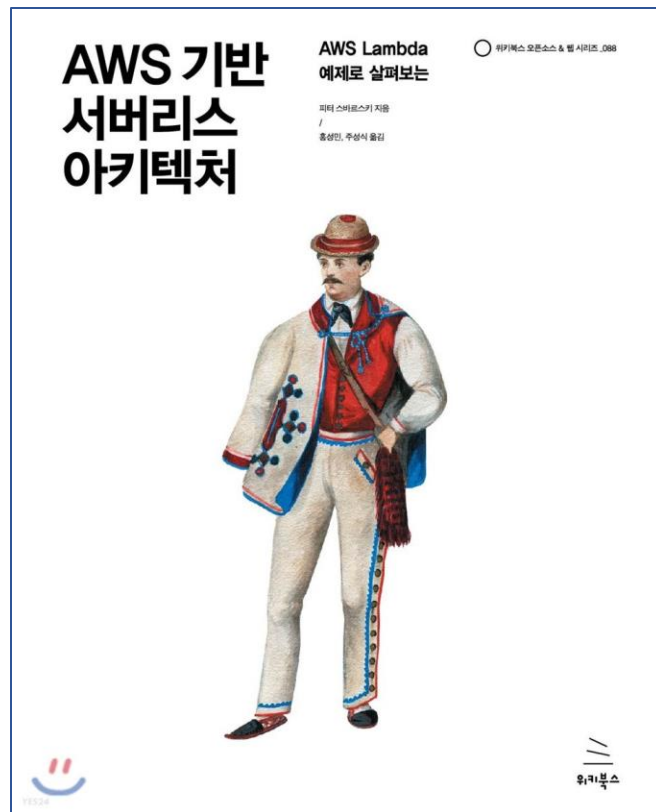


Textbook

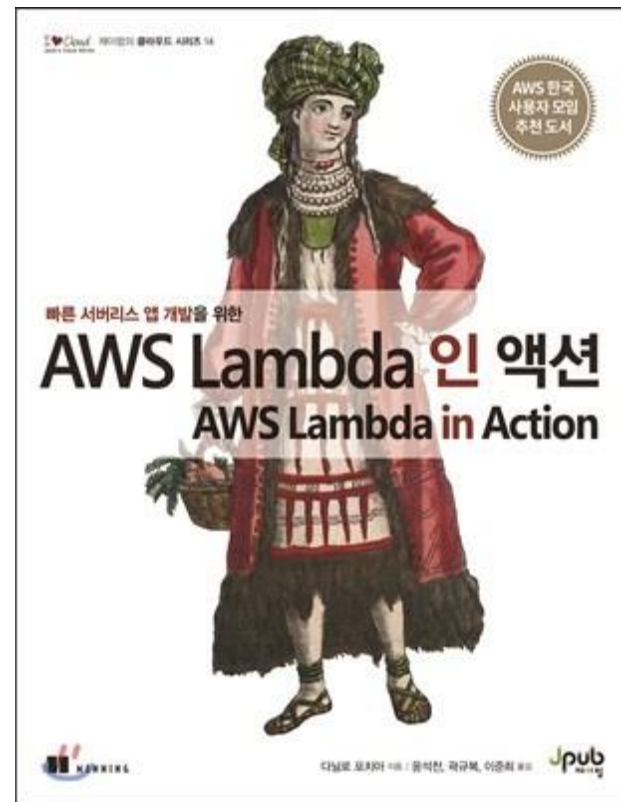
- Amazon AWS Documentation
 - <https://aws.amazon.com>
 - <https://docs.aws.amazon.com>



AWS 기반 서버리스 아키텍처

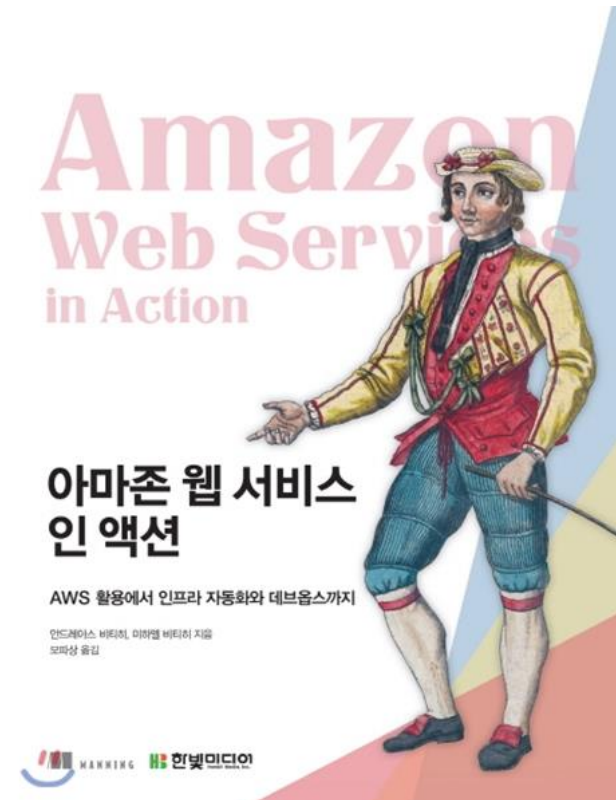


AWS Lambda 인 액션



Textbook

- **Inside JavaScript** (송현주, 고현준) 한빛미디어
- **아마존 웹 서비스 인 액션** (모파상) 한빛미디어



Textbook

Youtube Contents

The screenshot shows a YouTube search results page for the query 'amazon aws tutorial'. The left sidebar contains navigation icons for Home, Library, History, and various playlists. The main content area displays a list of video results. The top result is 'Getting Started with AWS | Amazon Web Services BASICS' by Academind, with 44,000 views and a duration of 2:01. Below it is 'AWS Tutorial For Beginners | AWS Certified Solutions Architect | AWS Training | Edureka' by edureka!, with 1,490,000 views and a duration of 2:00:22. Other results include 'AWS In 10 Minutes | AWS Tutorial For Beginners | AWS Training Video | AWS Tutorial | Simplilearn', 'AWS Tutorial For Beginners | AWS Training | Intellipaat', 'AWS Technical Essentials Training', and 'Amazon AWS Tutorial #1: Cloud Sysadmin - Your First EC2 Instance'.

Web Contents

The screenshot shows a Google search results page for the query 'amazon aws tutorial'. The search bar at the top contains the text 'amazon aws tutorial'. Below the search bar, the first result is 'platform that enables users to access on demand ...'. The second result is 'AWS Tutorial | Amazon Web Services Tutorial - Javatpoint' with the URL 'https://www.javatpoint.com › aws-tutorial'. Below this is another result for 'AWS Tutorial | Amazon Web Services Tutorial with aws, tutorial, introduction, amazon web services, aws history, features of aws, aws global infrastructure, aws ...'.

[AWS Tutorial | Amazon Web Services Tutorial - Javatpoint](https://www.javatpoint.com › aws-tutorial)

<https://www.javatpoint.com › aws-tutorial> ▾ 이 페이지 번역하기

AWS Tutorial | Amazon Web Services Tutorial with aws, tutorial, introduction, amazon web services, aws history, features of aws, aws global infrastructure, aws ...

관련 질문

How do I learn Amazon AWS?

Is Amazon Web Services easy to learn?

What is Amazon Web Services for beginners?

What is AWS technology?

사용자 의견

[AWS Tutorial: Introduction to Cloud Computing | Edureka Blog](https://www.edureka.co › blog › amazon-aws-tutorial)

<https://www.edureka.co › blog › amazon-aws-tutorial> ▾ 이 페이지 번역하기

2019. 5. 22. - aws logo - amazon aws tutorial - edureka. Amazon Web Services(AWS) is a Amazon.com subsidiary which offers cloud-computing services at ...

[Free Amazon AWS Tutorial - Learn Amazon Web Services \(AWS\): The ...](https://www.udemy.com › course › learn-amazon-web-...)

<https://www.udemy.com › course › learn-amazon-web-...> ▾ 이 페이지 번역하기

Start your journey with AWS, get hands on, and cover every AWS service available today - Free Course.

[Introduction to AWS | AWS Tutorial for Beginners | A Cloud Guru](https://acloud.guru › learn › aws-technical-essentials)

<https://acloud.guru › learn › aws-technical-essentials> ▾ 이 페이지 번역하기

New to Amazon Web Services (AWS)? This 1 hour Amazon Web Services tutorial for beginners teaches you everything that you need to know about the "how"



Software

- **JavaScript, Node.JS, Express.JS, Axios.JS**
 - <https://nodejs.org/ko>
- **Visual Studio Code**
 - <https://code.visualstudio.com>
- **Postman**
 - <https://www.getpostman.com/>



Classes

- **Lectures**

- Pre-recorded Video on ecampus
(<http://ecampus.konkuk.ac.kr>)

- **Lab**

- Pre-recorded Video on ecampus
- Realtime Online Lab (zoom)

- **Real time Online Class (zoom) & In-Room Class**

- Project Presentation & Evaluation
- Q&A

Evaluation

- Attendance: 10%
- Project: 90%
 - 6개의 과제로 구성
 - 사례조사: 2개 과제
 - ✓ 스마트시티 & 공공데이터포털 프로젝트 사례 조사
 - ✓ Amazon AWS Cloud 프로젝트 사례 조사
 - 시스템 개발: 2개 (4개 과제)
 - ✓ 스마트시티 어플리케이션 기획
 - ✓ 스마트시티 어플리케이션 개발 (On-Premise 버전)
 - ✓ 스마트시티 어플리케이션 설계 (AWS 버전)
 - ✓ 스마트시티 어플리케이션 개발 (AWS 버전)

